# ER Diagram to Relational Database Schema

Conversion of Department – Course – Room – Class ER model (1–N relationships)

## Analysis

Entities: **Department**(DepartmentID, DepartmentName, Office), **Course**(CourseID, CourseName, Credits), **Room**(RoomID, RoomName, Capacity), **Class**(ClassID, Semester, Year, Schedule). Relationships **Offers** (Department 1 – N Class), **Has** (Course 1 – N Class), **ScheduledIn** (Room 1 – N Class) are all 1-to-N with Class on the N-side, so they collapse into foreign keys in Class. No junction tables are needed.

## 1. Relations and Attributes

| Relation | Attributes (PK = **bold**, FK = * ) |
|---|---|
| Department | **DepartmentID**, DepartmentName, Office |
| Course | **CourseID**, CourseName, Credits |
| Room | **RoomID**, RoomName, Capacity |
| Class | **ClassID**, Semester, Year, Schedule, DepartmentID*, CourseID*, RoomID* |

## 2. Primary Keys (PK)

Department(**DepartmentID**)
Course(**CourseID**)
Room(**RoomID**)
Class(**ClassID**)

## 3. Foreign Keys (FK)

| FK | References |
|---|---|
| Class.DepartmentID | Department(DepartmentID) — via Offers |
| Class.CourseID | Course(CourseID) — via Has |
| Class.RoomID | Room(RoomID) — via ScheduledIn |

All three FKs in Class are NOT NULL — every class must belong to one department, one course, and one room.

## 4. Referential Integrity Constraints

Class[DepartmentID] $\subseteq$ Department[DepartmentID]
Class[CourseID] $\subseteq$ Course[CourseID]
Class[RoomID] $\subseteq$ Room[RoomID]

### Relational Schema Notation

Department(**DepartmentID**, DepartmentName, Office)
Course(**CourseID**, CourseName, Credits)

Room(**RoomID**, RoomName, Capacity)
Class(**ClassID**, Semester, Year, Schedule, *DepartmentID, CourseID, RoomID*)

## SQL DDL

```
CREATE TABLE Department(
  DepartmentID INT PRIMARY KEY,
  DepartmentName VARCHAR(100),
  Office VARCHAR(100)
);
CREATE TABLE Course(
  CourseID INT PRIMARY KEY,
  CourseName VARCHAR(100),
  Credits INT
);
CREATE TABLE Room(
  RoomID INT PRIMARY KEY,
  RoomName VARCHAR(100),
  Capacity INT
);
CREATE TABLE Class(
  ClassID INT PRIMARY KEY,
  Semester VARCHAR(20),
  Year INT,
  Schedule VARCHAR(100),
  DepartmentID INT NOT NULL REFERENCES Department(DepartmentID),
  CourseID INT NOT NULL REFERENCES Course(CourseID),
  RoomID INT NOT NULL REFERENCES Room(RoomID)
);
```